

```
#elif defined(__PCH__) // same as above but for the PCH
compiler and PIC18F452
#include <18F452.h>
#fuses HS,NOWDT,NOPROTECT,NOLVP
#use delay(clock=20000000)
#use rs232(baud=9600, xmit=PIN_C6, rcv=PIN_C7)
#endif

void main() { // main function
    inti, value, min, max; // local variable declaration
    printf("Sampling:"); // printf function included in the
    RS232 library
    setup_port_a( ALL_ANALOG ); // A/D functions- built-in
    setup_adc( ADC_CLOCK_INTERNAL ); // A/D functions-
    built-in
    set_adc_channel( 0 ); // A/D setup functions- built-in
    do { // do while statement
        min=255; // expression
        max=0;
        for(i=0; i<=30; ++i) { // for statement
            delay_ms(100); // delay built-in function call
            value = Read_ADC(); // A/D read functions- built-in
            if(value<min) // if statement
                min=value;
            if(value>max) // if statement
                max=value;
        }
        printf("\n\rMin: %2X Max: %2X\n\r",min,max);
    } while (TRUE);
}
```

This code can be only compiled by CCS compilers since it uses CCS specific functions and pre-processor directives. Though it may look daunting, after you finish with this section, you'll be able to understand it.

4.1 Concepts

This part covers some basic concepts of the language, essential for understanding how C programs work. First, we need to establish the following terms that will be used throughout the help: